

Consumer Buying and Selling

JEB104 Microeconomics I
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Prague, IES FSV UK, April 20, 2020

Buying and Selling

Previously

- In most of our previous lectures, we assumed that consumer's income is monetary and it is given.
- The only exception was the expenditure minimization problem and derivation of Hicksian demand (consumer aims to reach a given utility level with minimum spending).

Now

- In reality, people earn their income by selling what they own, what they produced, assets they have accumulated, or most commonly their own labor.
- We will modify the baseline model to describe this behavior.
- $\Rightarrow$ consumer can buy goods (as in previous analysis), but also sell goods.

If you want to buy something, you need to give something else away.

1. Which goods will consumer and which goods will consumer buy?
2. How preferences of a consumer look like, if he is buyer or seller of some good?
3. If income is generated by selling what consumer owns, what will be its size?
4. Do price changes affect whether consumer buys or sells the good? How?

Trade involves exchange: something bought $\Leftrightarrow$ something sold

Questions to answer:

1. What will be bought and what will be sold?
2. Who will be the buyer and who will be the seller?
3. How is income generated?
4. How do price changes affect demand?

Assumptions

Endowment

- Consumer has some initial endowment of goods $(\omega_1, \omega_2, \dots, \omega_n)$, $\omega_i \geq 0$.
- Endowment's value (wealth):

$$w = \sum_{i=1}^n p_i \omega_i$$

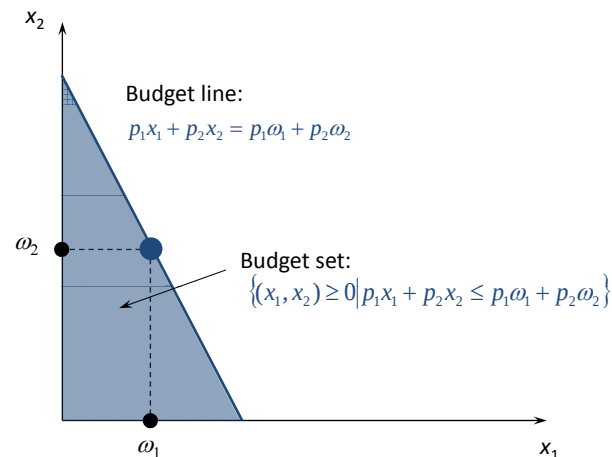
Budget constraint:

$$\sum_{i=1}^n p_i x_i \leq \sum_{i=1}^n p_i \omega_i = w$$

Budget set:

$$B = \{x \in \mathbb{R}_{+,0}^n : \sum_{i=1}^n p_i x_i \leq w\}$$

Budget Constraint



Endowment

- defines how much of the goods the consumer has before he enters the market (he produced them, or he just owns them)
- $(\omega_1, \omega_2) = (10, 2)$ states that the consumer is endowed with 10 units of good 1 and 2 units of good 2, consider a farmer who produced 10 units of corn and 2 units of potatoes (1 unit can be 1 bushel for example, in the case of corn, it is around 25kg)

Wealth

- Value of the endowment: market prices times endowment levels
- We assume there is no outside monetary income (for simplicity)
- under the market prices, you can sell your whole endowment and get income equal to wealth w
- Farmer: potatoes and corn, if $p_1 = 20$ and $p_2 = 30$, then $w = p_1 \omega_1 + p_2 \omega_2 = 20 \cdot 10 + 30 \cdot 2 = 260$

Budget constraint and budget set

- For which bundles the initial endowment can be exchanged – any bundle x costing no more than the value of endowment w
- spending on goods consumer consumes $\leq$ value of endowment
- imagine that the farmer sells all his endowment, gets income 260 and we are back in our initial analysis (so budget constraint of this farmer before selling his endowment is the same as the budget constraint he is facing if he has sold his entire endowment and received income 260)
- Slope is ratio of prices as in the baseline analysis
- But the budget line has to go THROUGH the endowment point!

Budget constraint graphically

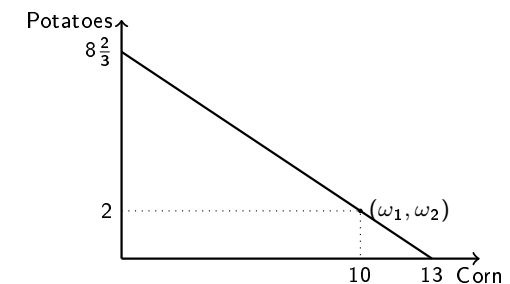
- 2-goods case – rather limited, assumption that we consume goods which we own (farmer grows corn and potato and consumes these 2 goods)
- Endowment point (ω_1, ω_2) on budget line – consumer may ALWAYS keep her endowment and consume it

Intersection points:

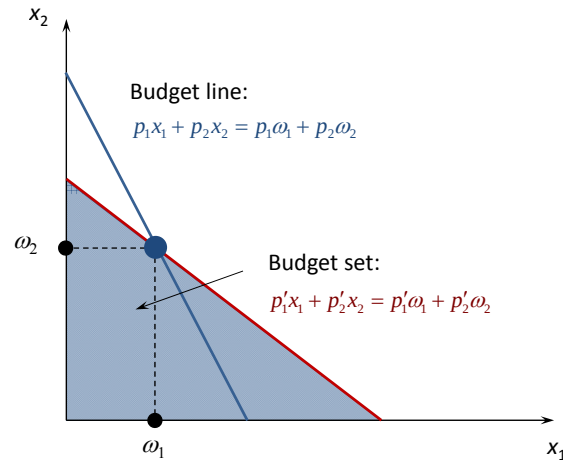
- $x_2 = 0 \Rightarrow x_1 = \omega_1 + \frac{p_2 \omega_2}{p_1}$
- $x_1 = 0 \Rightarrow x_2 = \omega_2 + \frac{p_1 \omega_1}{p_2}$
- Both are “initial endowment plus what I get if I sell all of the other good's endowment”

Corn and potatoes example ($p_1 = 20$ and $p_2 = 30$).

Only corn: $x_1 = 10 + \frac{30 \cdot 2}{20} = 13$, only potatoes: $x_2 = 2 + \frac{20 \cdot 10}{30} = 8 \frac{2}{3}$

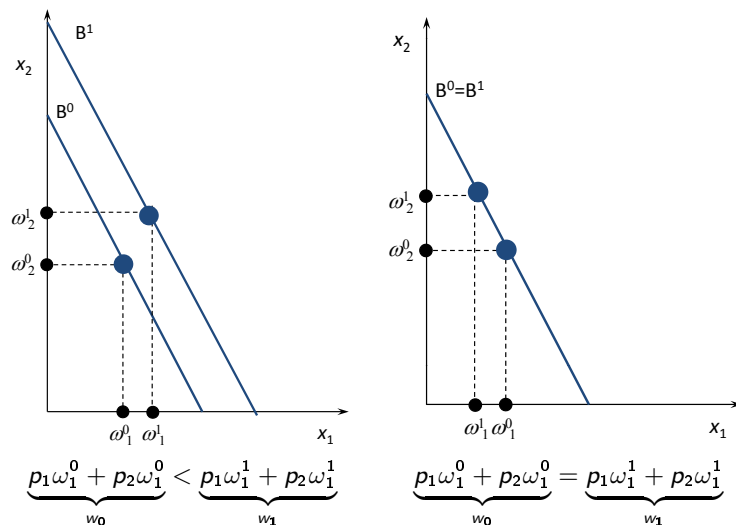


Budget Constraint: Change in Prices



- The endowment point is always on the budget constraint.
- Price changes pivot the constraint around the endowment point.

Budget Constraint: Change in Endowment



Old budget set given by blue budget line, new budget set given by red budget line and shaded area.

Budget line flatter under new prices, so what happened with prices?

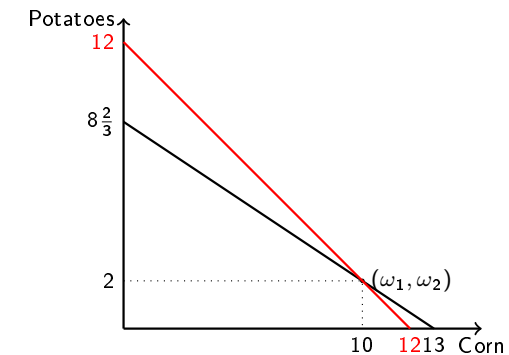
- slope is $-\frac{p_1}{p_2}$, budget line can be expressed by this formula: $x_2 = -\frac{p_1}{p_2}x_1 + \frac{w}{p_2}$, where $w = p_1\omega_1 + p_2\omega_2$
- Figure shows: $\frac{p_1}{p_2} > \frac{p'_1}{p'_2}$ (new budget line flatter)
- Consumer could experienced:
 - decrease of p_1 , increase of p_2 , or both
 - greater decrease of p_1 than of p_2 , or greater increase of p_2 than of p_1

Imagine different price changes

- Proportional increase in all prices – no change as wealth changes with the same proportion as prices, budget line always passes through the endowment point
- Change only in one price (budget line rotates around the endowment point)
- Change in both prices (budget line also rotates around the endowment point)

Farmer's example: change in price of corn from $p_1^0 = 20$ to $p_1^1 = 30$ ($p_2 = 30$)

Only corn: $x_1 = 10 + \frac{30 \cdot 2}{30} = 12$, only potatoes: $x_2 = 2 + \frac{30 \cdot 10}{30} = 12$



Increasing price of corn means that on one hand, farmer can afford to buy more potatoes, because he gets higher revenues from selling his endowment of corn, on the other hand, he can afford less corn if he consumes more than what is his initial endowment.

Left figure: New budget line further from the origin than the initial budget line

- Consumer is endowed with more of both goods (as shown in the figure)
- Consumer could be also endowed with much more of one and a little less of the other (so $\omega_1^0 > \omega_1^1$ and $\omega_2^0 < \omega_2^1$ and $w^0 < w^1$)

Right figure: Endowment changes, but the budget constraint stays the same.

- Necessarily more of one good and less of the other, value of the endowment does not change.

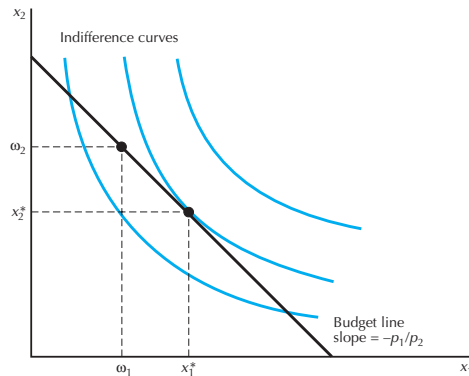
Optimum and Gross Demand

Optimization:

$$\max u(x_1, x_2, \dots, x_n), \quad \text{s.t.} \quad \sum_{i=1}^n p_i x_i \leq w, \quad x_i \geq 0 \quad \forall i \in \{1, 2, \dots, n\}$$

Interior optimum:

$$MRS_{ji} = -\frac{p_i}{p_j} \rightarrow x_i^*(p_1, \dots, p_n, \omega_1, \dots, \omega_n) = x_i^*(p, \omega) = D_i(p, \omega)$$



Gross demand for good i : $x_i^*(p, \omega) = D_i(p, \omega)$

Net Demand

Net demand $\hat{D}_i(p, \omega)$: $\hat{x}_i^* = x_i^* - \omega_i$

- **Net buyer** of good i : $\hat{D}_i(p, \omega) = x_i^* - \omega_i > 0$
- **Net seller** of good i : $\hat{D}_i(p, \omega) = x_i^* - \omega_i < 0$

In optimum:

$$\sum_{i=1}^n p_i (x_i^* - \omega_i) = 0$$

Value of what the consumer buys = value of what the consumer sells

We can use a standard approach to find consumer's optimum, so we max utility subject to budget constraint. There is no substantial difference to the standard case, except that now the endowment (ω) matters and affects demand instead of monetary income.

Optimality condition holds in the interior optimum. From the utility max we derive so called Gross demands for good i (showing how much of good i consumer wants to consume). This is the same for given budget constraint irrespective of the level of consumer's endowment (it does not matter where on a particular budget line endowment point lies).

In optimum (x_1^*, x_2^*) shown in the figure,

- consumer consumes less of ω_2 , i.e. $\omega_2 > x_2^*$, so consumer is selling good 2
- consumer consumes more of ω_1 , i.e. $\omega_1 > x_1^*$, so consumer is buying good 1.

Net demands tell us whether consumer buys or sells the good.

Net demands are Gross demands minus values of endowments, so they can be even negative.

In optimum

- Rewritten budget constraint when it holds with equality in optimum:

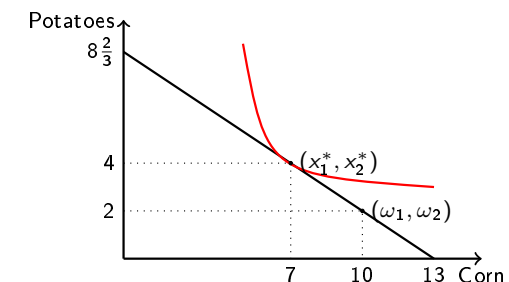
$$\sum_{i=1}^n p_i x_i^* = \sum_{i=1}^n p_i \omega_i \Leftrightarrow \sum_{i=1}^n p_i (x_i^* - \omega_i) = 0$$

- Sum of the values of consumer's net demands is zero.

Value of purchases equals value of sales

- You sell some endowment and get some income out of it, then you use this income to buy the other good.
- If consumer earns 100 of dollars from sales of one good, he can use just these 100 dollars for purchases of other goods.

Farmer's example: Gross demand $x^* = (7, 4)$



Farmer is net buyer of potatoes: $\hat{x}_2^* = x_2^* - \omega_2 = 4 - 2 = 2 > 0$

Farmer is net seller of corn: $\hat{x}_1^* = x_1^* - \omega_1 = 7 - 10 = -3 < 0$

Change in Endowment

Change in endowment, given prices, such that:

B^1 above B^0 (wealth increases):

- gross demand for good i increases if it is normal good
- gross demand for good i decreases if it is inferior good

B^1 below B^0 (wealth decreases):

- gross demand for good i decreases if it is normal good
- gross demand for good i increases if it is inferior good

B^1 identical B^0 (wealth does not change):

- gross demand for good i does not change; only net demand changes

Change in Prices

Change in prices $\Rightarrow$ change in wealth

Before change of price of good i :

- consumer is net buyer of good i
- consumer is net seller of good i

After change of price of good i :

- original net buyer continues to be net buyer of good i
- original net buyer switches to be net seller of good i
- original net seller continues to be net seller of good i
- original net seller switches to be net buyer of good i

We should discuss now how changes in endowment and prices affect demand.

For endowment changes, we just conclude what we already now (we distinguish between normal and inferior goods). It is impossible to say anything about net demands unless we know how exactly endowment changed.

In the baseline analysis: demand changes with price change given constant income

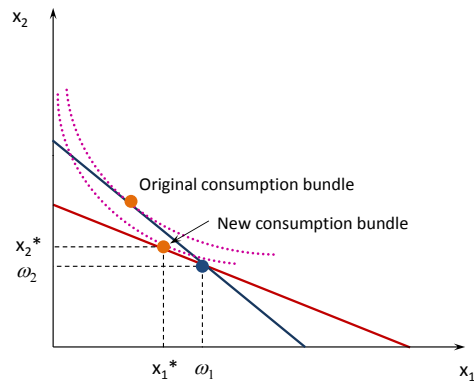
- But now with change in price there is a corresponding change in wealth
- Money income determined by the value of endowment, so you can imagine that with change in price also money which we have for consumption of other goods changes.

There are several options which we can analyze after a price change.

We will focus on a price decrease and analyze whether all these cases may occur.

Decrease of Price: Original Net Seller

Price of good 1 decreases: $|\frac{p_1}{p_2}|$ decreases $\rightarrow$ budget line flatter



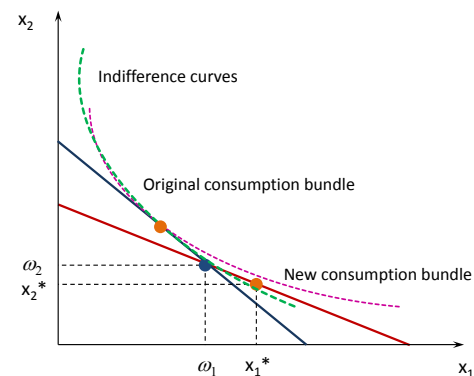
Original net seller of good 1 continues to be net seller of good 1.

Consumer must be worse off.

If the price of a good that a consumer is selling decreases and consumer decides to remain a seller, then he must be worse off.

Decrease of Price: Original Net Seller

Price of good 1 decreases:



Original net seller of good 1 switches to be net buyer of good 1.

Consumer may be worse (purple IC) or better (green IC) off.

If the price of a good that a consumer is selling decreases and consumer decides to start buying a good, then we cannot tell whether he is worse or better off.

If price of good 1 increases,

- consumer can afford more x_1 if he consumes more x_1 than his initial endowment; after the price change, he may sell some x_2 for the same price as before the price change of x_1 , but now he purchases x_1 for lower price.
- consumer can afford less x_2 if he consumes more x_2 than his initial endowment; after the price change, he sells some x_1 for lower price than before the price change, so his generated income is lower and can afford a lower amount of x_2 .

If consumer is a net seller ($x_1^* < \omega_1$), after the price decrease, his income generated from revenues of his sales is lower (if he remains seller), so he must be worse off.

From revealed preferences:

- The "left" part of the new budget line lies inside the original budget set
- Hence all of these choices were available also before the price change, original bundle was preferred to all these choices.
- $\Rightarrow$ Consumer (who remains net seller of x_1 and hence, his new optimum is located on this "left" part of the new budget line) is worse off after the price change

Here we have another option: original net seller of x_1 switches to be a net buyer after the price of good 1 decreases.

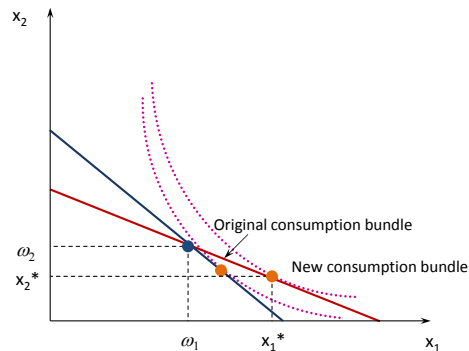
The original consumption point not available under the new prices, but the new consumption bundle not available under the old prices.

- Therefore we cannot tell whether consumer is better or worse off using revealed preferences.

The Figure shows two different indifference curves passing through the original consumption bundle. If consumer has the green indifference curve, he will be better off under new prices. If consumer has the purple indifference curve, he will be worse off under new prices. So both options are possible.

Decrease of Price: Original Net Buyer

Price of good 1 decreases:



Original net buyer of good 1 remains to be net buyer of good 1.
Consumer will be always better off.

*If the price of a good that a consumer is buying decreases, consumer will **always remain net buyer** and he will be better off.*

Increase of Price

Analogically, for price increase of a good:

- ① *If the price of a good that a consumer is buying increases and consumer decides to remain a buyer, then he must be worse off.*
- ② *If the price of a good that a consumer is buying increases and consumer decides to start selling the good, then we cannot tell whether he is worse or better off.*
- ③ *If the price of a good that a consumer is selling increases, consumer will **always remain a seller** and he will be better off.*

Here, we have the case of original net buyer. So suppose that consumer is a net buyer of x_1 and its price decreases.

Again from revealed preferences:

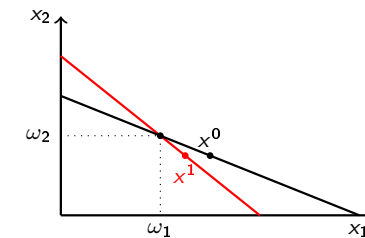
- Original consumption bundle is available after the price change, hence lying inside the new budget set.
- Hence new consumption bundle is directly revealed preferred to the original one.
- $\Rightarrow$ Consumer is better off

Can we have a situation that consumer switches to be net seller in the case of price decrease?

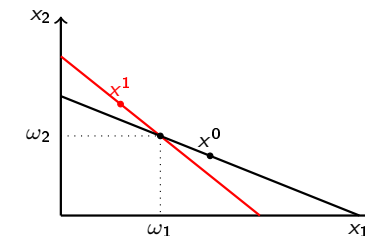
- Never, the consumer would violate WARP.
- New consumption bundle will never be located on the left part of the new budget line as these bundles were originally available but not chosen.

After the price increase of x_1 , budget line becomes steeper (red).

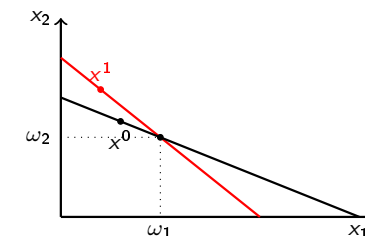
1. If consumer remains buyer after the price change, he must be worse off.



2. If consumer becomes net seller, we cannot assess whether he is better or worse off.



3. Net seller will always keep selling the good if its price increases.



Price-Offer Curve

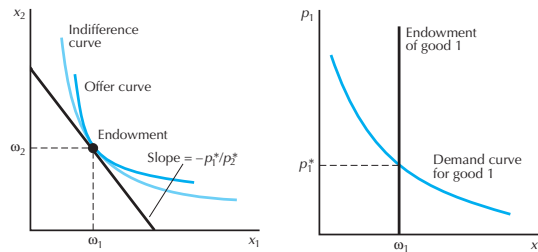
Connecting all the utility-maximizing gross demands for which the given endowment can be exchanged when price changes.

Price-offer curve passes through endowment for specific relative prices $\left| -\frac{p_1^*}{p_2^*} \right|$.

Suppose that price of good 1 changes:

If $\left| -\frac{p_1}{p_2} \right| > \left| -\frac{p_1^*}{p_2^*} \right|$: consumer becomes net seller of x_1 and net buyer of x_2

If $\left| -\frac{p_1}{p_2} \right| < \left| -\frac{p_1^*}{p_2^*} \right|$: consumer becomes net buyer of x_1 and net seller of x_2



Slutsky Equation Revisited

Original Slutsky equation:

- how demand reacts to a change in price, given constant money income

In this case: value of endowment changes with price change

Slutsky equation for own-price change revisited:

$$\frac{\partial D_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{\text{Substitution effect}} - \underbrace{D_i \frac{\partial D_i}{\partial m}}_{\text{Income effect}} + \underbrace{\omega_i \frac{\partial D_i}{\partial m}}_{\text{Endowment income effect}}$$

For one specific relative price ratio, endowment point maximizes utility, i.e. indifference point touches budget line exactly in the endowment point. In such case, consumer keeps consumes his endowment and there is no trade. To understand why this is the case, you can do the following:

- Draw an arbitrary endowment point.
- There must be an indifference curve passing through the endowment point, so draw this indifference curve.
- Now try to draw a tangency line to the indifference point in the endowment point, this tangency line can be our budget line which corresponds to relative prices, under which endowment point is optimum.

If p_1 increases, then the budget line will be steeper, but still passing through the endowment point. Consumer could afford the endowment point, but he prefers to sell some amount of good 1 which shifts him to higher indifference curve (so price-offer curve lies above the original indifference curve that goes through the endowment point).

If p_1 decreases, then the budget line will be flatter, but still passing through the endowment point. Consumer could afford the endowment point, but he prefers to buy some additional amount of good 1 which shifts him to higher indifference curve (so price-offer curve lies above the original indifference curve that goes through the endowment point).

Right picture shows Gross demand.

- Net demand can be obtained just by moving the vertical axis to the endowment ω_1
- Net seller to the left of the endowment ω_1 , net buyer to the right of the endowment ω_1

Above applications of revealed preference are handy, but they do not really answer the main question. Offer curve may look differently for different preferences.

We know that with increase of price, net seller keeps selling the good, but is he selling more (generates higher income) or less (because he is wealthier and wants to increase consumption)? With decrease of price, net buyer keeps buying the good, but is he buying more (good is cheaper) or less (lower income)?

How does the demand for a good react to a change in its price?

- Previous case, if money income constant and good normal – reduction in its price must lead to an increase in demand.
- Now, money income cannot be constant since the value of the endowment will change when a price changes.

Slutsky equation looks similar as original Slutsky equation, however income effect contains also effect of initial endowment, so besides substitution effect and income effect as in previous Slutsky equation, we have in addition Endowment (income) effect

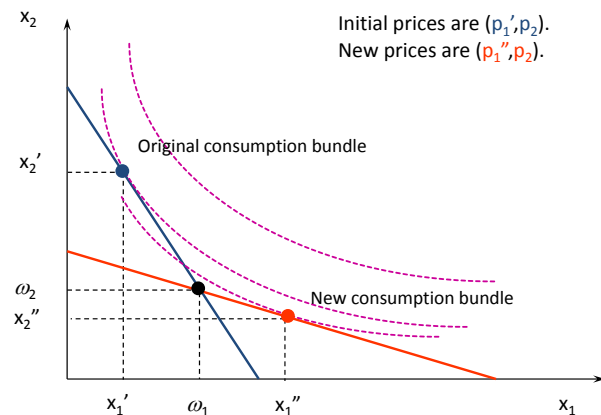
Income effect (that we already know) and endowment effect have opposite signs, why?

Suppose price increases:

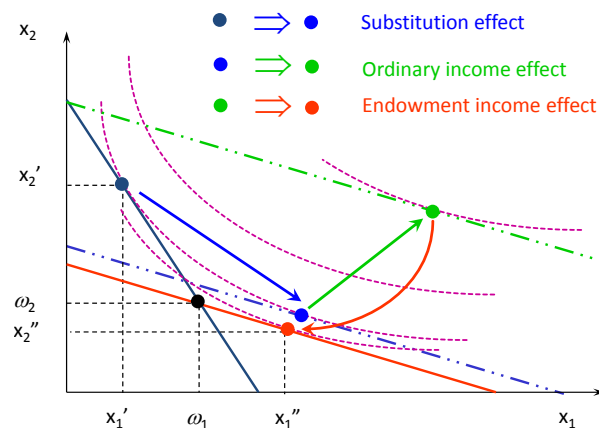
- Income effect shows the effect of the decrease in consumer's real income (and the decrease in real income is more severe, more of the good consumer demands). With the decrease of real income, consumer gets poorer and demands less of a normal good and more of an inferior good.
- Endowment effect however shows the effect of increased value of consumer's endowment. With price increase, consumer becomes richer with higher endowment of this good (note that his wealth increases, as he can sell his endowment for a higher price). If he is richer, he demands more of a normal good and less of an inferior good.
- So these two effects work in opposite direction and their sizes depend on how much consumer demands and how much consumer is endowed.

Slutsky Equation Revisited

Price of good 1 decreases:



Slutsky Equation Revisited



Movement of demand due to endowment effect = same sign as price change (x_i normal good)

Now we will try to decompose the total change in demand to the substitution effect, the income effect and the endowment effect.

We study the situation when price of good 1 decreases.

- In the original consumption bundle, the consumer is selling good 1 and buying good 2.
- After the price decreases, the consumer starts buying good 1 and selling good 2.
- Looking at indifference curves depicted in the figure, we can conclude that the consumer is worse off after the price decrease.
- Utility decreased even though price decreased, because the endowment income effect devaluates the income gains from price decrease. As the consumer was initially selling the good, decrease in price of this goods decreases his wealth.

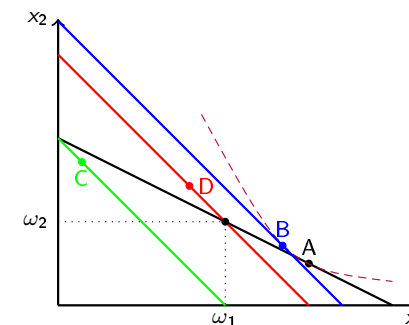
The blue movement shows the classical Hicksian substitution effect – so we move along the original indifference and aim to find a point where the original indifference curve is touching a tangency line having slope of new price ratio.

The green point shows what would be the optimum if consumer's income is given and does not change with the price change (so the green point would be our new optimum in a standard analysis). So the green movement is our standard income effect.

The red movement shows the endowment income effect. If price decreases, our income that is generated by a value of our endowment decreases, too. Because the good 1 is a normal good, the endowment effect works in the same direction as the price change (decrease of price $\rightarrow$ decrease of good demanded).

The picture below illustrates the decomposition to SE, IE and EE for a price increase.

- Original optimum – point A.
- Price of good 1 increases, hence budget line becomes steeper, new optimum is point D. Let's decompose total change in demand from the point A to the point D into separate effects.
- Substitution effect – movement from the point A to the point B along original indifference curve.
- Income effect – movement from the point B to the point C.
- Endowment income effect – movement from the point C to the point D.
- In this example, x_1 is a normal good, x_2 is an inferior good.



Slutsky Equation Revisited: Own Price Change

$$\frac{\partial D_i}{\partial p_i} = \underbrace{\frac{\partial h_i}{\partial p_i}}_{<0} - (D_i - \omega_i) \frac{\partial D_i}{\partial m}$$

$\frac{\partial D_i}{\partial p_i}$	$D_i - \omega_i$	$\frac{\partial D_i}{\partial m}$	
—	+	+	
—	—	—	
—	+	—	$ SE > IE + EE $
—	—	+	$ SE > IE + EE $
+	+	—	$ SE < IE + EE $
+	—	+	$ SE < IE + EE $

- Net buyer of a normal good: $\uparrow$ demand with $\downarrow$ price
- Net seller of an inferior good: $\downarrow$ demand with $\uparrow$ price
- All other cases: sign depends on size of income and substitution effects

Own substitution effect is always negative as with the standard Hicksian decomposition.

We have to take into account whether customer is a net buyer or a net seller.

Legend:

- $D_i - \omega_i$: + for net buyer, — for net seller
- $\frac{\partial D_i}{\partial m} > 0$ for normal goods, < 0 for inferior goods.

Net buyer of a normal good: $\uparrow$ demand with $\downarrow$ price

- if price of x a consumer is buying decreases and x is a normal good, he will buy more of this good. With price decrease, real income of the consumer increases and hence his demand for a normal good increases, too.

Net seller of an inferior good: $\downarrow$ demand with $\uparrow$ price

- if price of x a consumer is selling increases and x is an inferior good, he will sell more of this good. With price increase, the consumer gets higher revenues from his sales, so he becomes richer. As he is richer, he demands less of this inferior good (i.e. sells more of this good).